

JEE Main - 3 | JEE 2022 (Gen 1)

Date: 21/09/2020

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 999 (both inclusive).
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

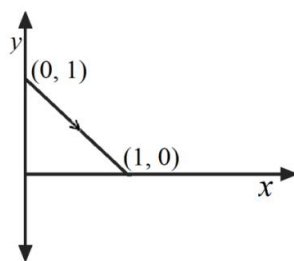
PART I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. A particle of mass m is tied to a massless string and the other end of the string is fixed at a point. Now, the particle is allowed to move in a circular path in the horizontal plane at a constant angular velocity such that the string makes a constant angle θ with the vertical. The tension in the string is:

(A) $mg \cos \theta$ (B) $mg \sin \theta \cos \theta$ (C) $\frac{mg}{\cos \theta}$ (D) $\frac{mg}{\sin \theta \cos \theta}$

2. A particle moves from the point $(0,1)$ to the point $(1,0)$ along a straight line as shown. If one of the forces acting on the particle was $F = 10 \text{ N}$, directed towards the $+X$ direction, the work done by F is equal to:



(A) Zero (B) 10 J (C) -10 J (D) 5 J

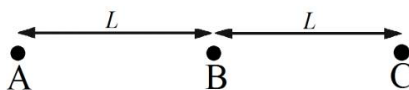
3. A ball is released from rest from a height h_0 . After rebounding once from the ground, it rises to a maximum height h_1 (such that $h_1 < h_0$). Then, after rebounding once more from the ground, it rises to a maximum height:

(A) $\sqrt{\frac{h_1^3}{h_0}}$ (B) $\sqrt{\frac{h_0^3}{h_1}}$ (C) $\frac{h_1^2}{h_0}$ (D) $\frac{h_0^2}{h_1}$

4. A block of mass 0.5 kg is placed on a fixed plane inclined at an angle $\theta = \sin^{-1}(0.6)$ with the horizontal. The coefficient of friction between the block and the plane is 0.5. The block starts moving down the inclined plane, starting from rest. The velocity of the block after it has moved a distance 1m is: ($g = 10 \text{ m/s}^2$)

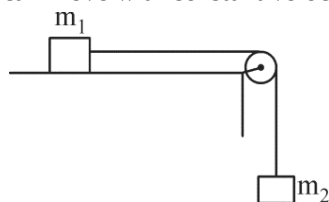
(A) 1 m/s (B) 1.5 m/s (C) 1.8 m/s (D) 2 m/s

5. Three particles A, B and C of mass m , $2m$ and $4m$ respectively are placed in a straight line as shown. The distance of the centre of mass of the system of the three particles from A and B respectively is:

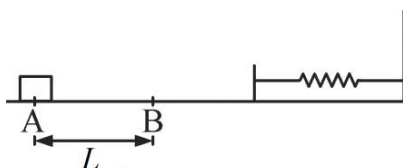


(A) $\frac{10L}{7}, \frac{3L}{7}$ (B) $\frac{9L}{7}, \frac{2L}{7}$ (C) $\frac{4L}{7}, \frac{3L}{7}$ (D) $\frac{5L}{7}, \frac{2L}{7}$

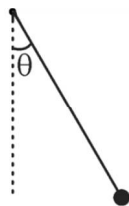
6. Two blocks of masses m_1 and m_2 are arranged with an ideal pulley and a massless string as shown, with the block of mass m_1 on a rough horizontal table. The required coefficient of friction between the block and the table such that the blocks can move with constant velocity is:



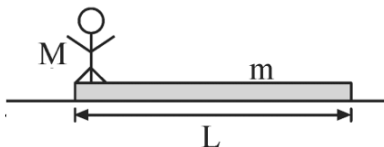
- (A) $\frac{m_1}{m_2}$ (B) $\frac{m_2}{m_1}$ (C) $1 - \frac{m_1}{m_2}$ (D) $\frac{m_1 - m_2}{m_1 + m_2}$
7. A block of mass m rests on a horizontal surface and an ideal spring of spring constant k is connected to a rigid wall as shown. The spring is unstretched. The portion AB (of length L) of the surface is rough with coefficient of friction μ and the rest of the surface is frictionless. The block is projected from A with a velocity u towards the right. The minimum value of u such that the block strikes the spring once and then is able to reach the point A again is $u_{\min}$. Then, the value of $u_{\min}$ is:



- (A) Independent of m and k but dependent on μ and L
 (B) Independent of m but dependent on k , μ and L
 (C) Independent of m , L and k but dependent on μ
 (D) Dependent on m , k , μ and L
8. A small particle of mass m is tied to a massless thread and the other end of the thread is fixed at a point. The particle is released from rest with the string being tight and making an angle $\theta = 45^\circ$ with the vertical as shown. The acceleration of the particle when it is at the lowest point in its path is:

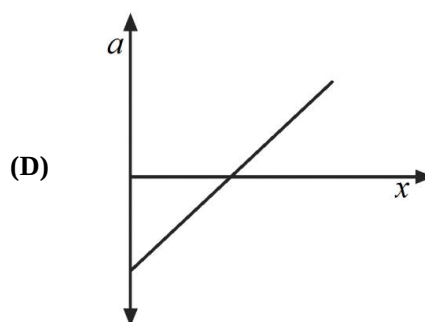
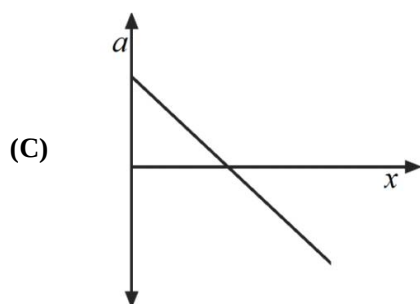
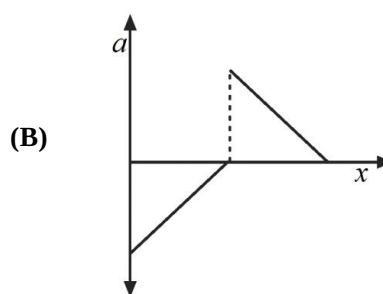
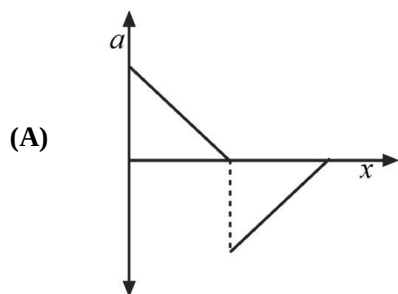
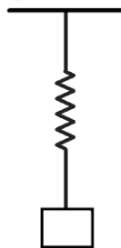


- (A) $(3 - \sqrt{2})g$ (B) $\frac{g}{\sqrt{2}}$ (C) $(2\sqrt{2} - 1)g$ (D) $(2 - \sqrt{2})g$
9. A man of mass M stands at one end of a plank of length L of mass m that rests on a smooth horizontal surface. If the man walks to the other end of the plank, the plank displaces by:

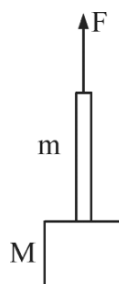


- (A) $\left(\frac{m}{M - m}\right)L$, towards right (B) $\left(\frac{m}{M - m}\right)L$, towards left
 (C) $\left(\frac{M}{M + m}\right)L$, towards right (D) $\left(\frac{M}{M + m}\right)L$, towards left

10. A block is suspended vertically by an ideal spring as shown, and the upper end of the spring is fixed. The block is released from rest when the spring is in its natural length. The graph that correctly depicts the acceleration of the block (a) as a function of the distance it has moved (x) until it comes to rest again is: (The vertically upward direction is taken positive)



11. A uniform rope of mass m is tied to a block of mass M and a constant force F is exerted on the rope in the vertically upward direction as shown. The force that the rope exerts on the block, if both move upwards with constant acceleration is:

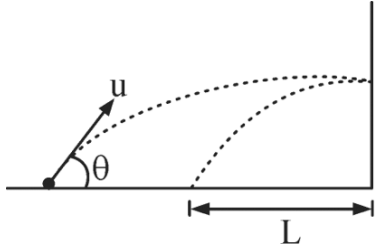


(A) $\left(\frac{M}{M+m}\right)F$

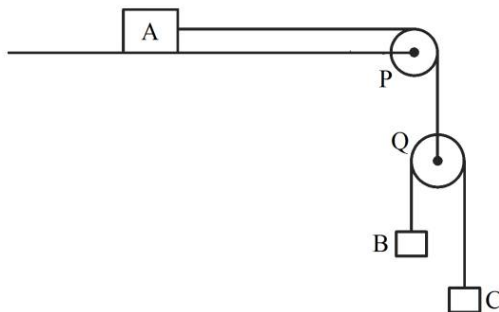
(B) $Mg + \left(\frac{M}{M+m}\right)F$

(C) $\left(\frac{M}{m}\right)F$

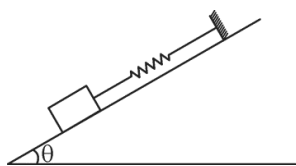
(D) $Mg + \left(\frac{M}{m}\right)F$

12. Two small blocks A and B of mass m_1 and m_2 respectively are placed on a circular disk that is fixed to rotate about its centre in the horizontal plane. The distance of the blocks from the centre of the disc is r_1 and r_2 respectively. The coefficient of friction between the blocks and the disk is μ_1 and μ_2 respectively. Starting from rest, the disk is rotated with about its centre with a constant angular acceleration. It is observed that block A slips on the disk before block B does. It can be concluded that:
- (A) $m_1\mu_1r_2 > m_2\mu_2r_1$ (B) $m_1\mu_1r_1 < m_2\mu_2r_2$ (C) $\mu_1r_2 < \mu_2r_1$ (D) $\mu_1r_2 > \mu_2r_1$
13. Two small disks A and B of mass m and $2m$ are moving on a smooth horizontal table with velocity $\vec{v}_A = 4\hat{i}$ and $\vec{v}_B = \hat{i} - 3\hat{j}$ (The surface of the table is the X-Y plane). At $t = 0$, the disks collide at the origin. At $t = 2$ s, the coordinates of the centre of mass of the two particles are:
- (A) $(2, -2)$ (B) $(4, -2)$ (C) $(4, -4)$ (D) $(2, -4)$
14. A particle is moving in the X-Y plane. Its momentum as a function of time is given by: $\vec{p}(t) = A\cos(\omega t)\hat{i} + B\sin(\omega t)\hat{j}$. Here A , B and ω are constants. The net force acting on the particle is given by:
- (A) $\vec{F} = -A\omega\sin(\omega t)\hat{i} + B\omega\cos(\omega t)\hat{j}$ (B) $\vec{F} = A\omega\sin(\omega t)\hat{i} - B\omega\cos(\omega t)\hat{j}$
 (C) $\vec{F} = -A\omega\cos(\omega t)\hat{i} + B\omega\sin(\omega t)\hat{j}$ (D) $\vec{F} = A\omega\cos(\omega t)\hat{i} - B\omega\sin(\omega t)\hat{j}$
15. A small particle is projected from the ground with velocity u at an angle θ with the horizontal towards a smooth vertical wall as shown. The particle strikes the wall horizontally, and after rebounding, strikes the ground at a distance L from the wall. The angle with the horizontal made by the velocity of the particle just before it strikes the ground is:
- 
- (A) $\tan^{-1}\left(\frac{u^2\sin^2\theta}{gL}\right)$ (B) $\tan^{-1}\left(\frac{u^2\cos^2\theta}{2gL}\right)$
 (C) $\tan^{-1}\left(\frac{gL\sin\theta}{u^2}\right)$ (D) $\tan^{-1}\left(\frac{gL\cos\theta}{2u^2}\right)$
16. A particle of mass 1 kg is moving with constant velocity 35 m/s along a straight line. A force of magnitude 15 N that makes a constant angle $\theta = \sin^{-1}(0.8)$ with the particle's initial direction of motion acts on the particle for 5 seconds. The speed of the particle after the force stops acting is:
- (A) 120 m/s (B) 100 m/s (C) $45\sqrt{2}$ m/s (D) $60\sqrt{3}$ m/s

17. Three blocks A, B and C of mass 4 kg, 1 kg and 3 kg respectively are arranged as shown with two ideal pulleys P and Q and massless strings. The block A rests on a fixed, smooth horizontal surface. The pulley P is fixed. The system is released from rest. The acceleration of the block B is:



- (A) $\frac{g}{7}$, downwards (B) $\frac{3g}{7}$, downwards
(C) $\frac{g}{7}$, upwards (D) $\frac{3g}{7}$, upwards
18. At $t = 0$, three mutually perpendicular forces of the same magnitude F start acting on a particle of mass m , which was initially at rest. Then, the work done on the particle by any one of the forces until $t = t_0$ is:
- (A) $\frac{F^2 t_0^2}{m}$ (B) $\frac{F^2 t_0^2}{2m}$ (C) $\frac{\sqrt{3} F^2 t_0^2}{2m}$ (D) $\frac{F^2 t_0^2}{3m}$
19. The potential energy of a particle is given by: $U(x) = 16 + \frac{1}{4}(x - 2)^2$. Which of these options is correct?
- (A) $x = 2$ is a stable equilibrium (B) $x = 2$ is an unstable equilibrium
(C) $x = 4$ is a stable equilibrium (D) $x = 4$ is an unstable equilibrium
20. A block of mass of 2 kg is connected to an ideal spring of spring constant 50 N/m and placed on a fixed plane inclined at an angle $\theta = \sin^{-1}(0.6)$. The other end of the spring is fixed. The coefficient of friction between the block and the plane is 0.2. The spring is stretched by 10 cm from its natural length and the block is released from rest. The acceleration of the block immediately after it is released is:
($g = 10 \text{ m/s}^2$)



- (A) 0.7 m/s^2 , up the plane (B) 2.6 m/s^2 , up the plane
(C) 1.9 m/s^2 , down the plane (D) Zero

SPACE FOR ROUGH WORK

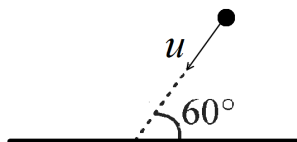
SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is an **integer** ranging from 0 to 999 (both inclusive).

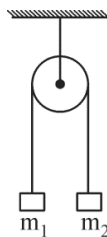
1. A particle of mass 1 kg is initially at rest. At $t = 0$, a time-varying force $F = 3t$ starts acting on the particle (t is time in seconds, and the force F is in Newton). The power being delivered by the force to the particle at $t = 2$ s is _____ W.
2. A block is travelling with velocity $u = 4\sqrt{2}$ m/s on a smooth horizontal surface towards a fixed smooth track of height $h = 0.8$ m that makes an angle 60° with the horizontal near its top. The velocity of the block at the highest point in its path is _____ m/s. ($g = 10 \text{ m/s}^2$)



3. Two particles of mass 1 kg and 3 kg are travelling directly towards each other on a smooth horizontal surface at speed 6 m/s and 8 m/s respectively. They have an elastic head-on collision. The speed of the particle of mass 1 kg after the collision is _____ m/s.
4. A ball of mass 120 g travelling with velocity $u = 10$ m/s at an angle 60° with the horizontal collides with a smooth horizontal surface. If immediately after the collision, the ball travels in a direction perpendicular to its initial direction, the kinetic energy lost by the ball during the collision is _____ J.



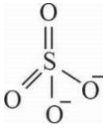
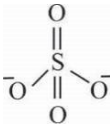
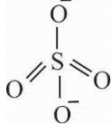
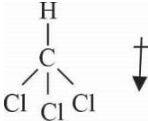
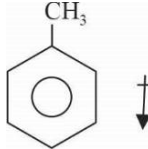
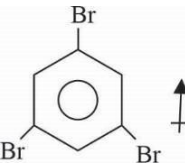
5. Two blocks of mass $m_1 = 600$ g and $m_2 = 400$ g are suspended using a massless string and an ideal pulley such that they are at the same horizontal level, and released from rest with the string being tight. The difference in height of the blocks 1 second after the start is _____ m. ($g = 10 \text{ m/s}^2$)



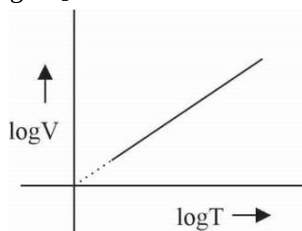
SPACE FOR ROUGH WORK

PART II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- The compound which contains only covalent bonds is:
 (A) KCN (B) NH_4Cl (C) $\text{PCl}_5(\text{solid})$ (D) H_2O_2
- A sigma bond may be formed by the overlap of 2 atomic orbitals of atoms A and B. If σ bond is formed along the x axis, which of the following overlaps is acceptable for π - bond?
 (A) s orbital of A and p_z orbital of B (B) p_x orbital of A and p_y orbital of B
 (C) p_z orbital of A and p_x orbital of B (D) p_y orbital of A and p_y orbital of B
- Out of $\text{CO}_2, \text{SO}_2, \text{NH}_3, \text{I}_3^-$ and I_3^+ , number of non-linear species are :
 (A) 1 (B) 3 (C) 2 (D) 4
- Choose the incorrect structure of SO_4^{2-}
 (A)  (B)  (C)  (D) $^- \text{O} - \text{O} - \text{S} - \text{O} - \text{O}^-$
- The longest C - C bond length is in:
 (A) C_6H_6 (B) C_2H_4 (C) C_2H_2 (D) C_2H_6
- Which one of the following compounds indicate wrong direction of dipole moment?
 (A)  (B)  (C)  (D) 
- The order of strength of hydrogen bonds is:
 (A) $\text{ClH} \cdots \text{Cl} > \text{NH} \cdots \text{N} > \text{OH} \cdots \text{O} > \text{FH} \cdots \text{F}$ (B) $\text{ClH} \cdots \text{Cl} < \text{NH} \cdots \text{N} < \text{OH} \cdots \text{O} < \text{FH} \cdots \text{F}$
 (C) $\text{ClH} \cdots \text{Cl} < \text{NH} \cdots \text{N} > \text{OH} \cdots \text{O} > \text{FH} \cdots \text{F}$ (D) $\text{ClH} \cdots \text{Cl} < \text{NH} \cdots \text{N} < \text{OH} \cdots \text{O} > \text{FH} \cdots \text{F}$
- Calculate formal charge on N atom in following compounds:
 (I) HNC (II) HCN (III) N_2 (IV) HNO_3
 (A) I = 0, II = 0, III = 0, IV = +1 (B) I = +1, II = 0, III = 0, IV = 0
 (C) I = +1, II = 0, III = 0, IV = +1 (D) I = +1, II = -3, III = 0, IV = +5
- Calculate density of CO_2 gas at 760 torr and 273°C .
 [Given: $R = 0.082 \text{ L atm Mol}^{-1}\text{K}^{-1}$, Atomic mass of C = 12 and O = 16]
 (A) 0.98g / L (B) 0.98g / ml (C) 1.96 g / L (D) $1.96 \times 10^{-3} \text{ g / L}$
- The temperature of an ideal gas is increased from 27°C to 327°C . Calculate ratio of de-broglie wavelength of ideal gas at 27°C and 327°C .
 (A) $\frac{2}{1}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{2}}{1}$ (D) $\frac{1}{1}$

11. At constant pressure of 0.1642 atm, $\log V$ vs $\log T$ graph is plotted for an ideal gas. Calculate moles of gas. [Given $R = 0.0821 \text{ L atm Mol}^{-1}\text{K}^{-1}$]



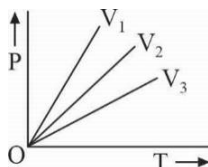
- (A) given graph is not possible (B) 2
(C) 3 (D) 4
12. The ratio between V_{rms} of CH_4 gas at 100 K and that of mixture of H_2 & CH_4 (molar ratio = 1:1) at 400 K [Given: Atomic Mass H = 1, C = 12]

- (A) $\frac{3}{8}$ (B) $\frac{1}{3}$ (C) $\frac{8}{3}$ (D) None of these

13. An air bubble of volume 1ml is at the bottom of river at 27°C and at 10 atm pressure. Calculate its volume at the top of the river if temperature at the top is 27°C .

- (A) 5 ml (B) 10ml (C) 15ml (D) 20ml

14. Under isochoric condition, graphs between P and T for an ideal gas are shown below. The correct order of volume is :



- (A) $V_1 = V_2 = V_3$ (B) $V_1 > V_2 > V_3$ (C) $V_1 < V_2 < V_3$ (D) $V_2 < V_1 < V_3$
15. Molar volume of an ideal gas is represented by:
- (A) 22.4L (B) 11.2L (C) $\frac{RT}{P}$ (D) 22.7L
16. Equal weights of H_2 , CH_4 and O_2 are mixed in a container at 127° . Calculate ratio of partial pressure of methane and total pressure. [Given Atomic Mass : H = 1, C = 12, O = 16]
- (A) $\frac{2}{19}$ (B) $\frac{1}{19}$ (C) $\frac{19}{38}$ (D) None of these
17. What volume of 0.3 M H_2SO_4 is required to exactly neutralize 200 ml of 0.5M NaOH ?
- (A) 1.8ml (B) 1.67 ml (C) 166.67 ml (D) 3.0L
18. How many nodal planes is/are present in molecular orbital $\pi^* 2p_x$?
- (A) 1 (B) 2 (C) 3 (D) 4
19. Calculate mass percentage of Nitrogen in 3.96g NH_4NO_3 . [Given Atomic Mass : H = 1, N = 14, O = 16]
- (A) 35% (B) 70% (C) 22% (D) 80%
20. In sp^3d^2 hybridization which orbitals are involved?
- (A) $d_{x^2-y^2}$ & d_{z^2} (B) d_{z^2} & d_{xy} (C) d_{xy} & d_{yz} (D) $d_{x^2-y^2}$ & d_{xy}

SPACE FOR ROUGH WORK

SECTION-2

This Section contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 999 (both inclusive).

- How many compounds have zero dipole moment?
 $\text{CO}_2, \text{SO}_2, \text{CCl}_4, \text{XeF}_4, \text{H}_2\text{O}, \text{CH}_2\text{Cl}_2, \text{CHCl}_3, \text{PCl}_5, \text{CH}_4, \text{BF}_3$
- Find the number of moles of BO formed in the given reaction if initial moles of AO = 10 mol, $\text{B}_2\text{O}_3 = 2$ mol, and yield of reaction is 25%.
 $\text{AO} + \text{B}_2\text{O}_3 \rightarrow \text{A}_2\text{O}_3 + \text{BO}$
- M_3X_8 is produced by the following steps of reactions:
Step – I : $\text{M} + \text{X}_2 \rightarrow \text{MX}_2$
Step – II : $\text{MX}_2 + \text{X}_2 \rightarrow \text{M}_3\text{X}_8$
If initially, M = 10 mol, $\text{X}_2 = 8$ mol. Then calculate moles of M_3X_8 formed
- 38g MgCl_2 is mixed with 23.4g of NaCl solution. Calculate total mole of ions
[Given: Atomic Mass of Na = 23, Mg = 24, Cl = 35.5]
- How many compounds will have produce two H^+ ions when dissolved in water?

H_3PO_3	H_3PO_4	H_2SO_4	H_2CO_3
HNO_3	CH_4	H_2O	H_3BO_3

SPACE FOR ROUGH WORK

PART III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- If the letters of word "VIRAT" are written in all possible ways and then are arranged in a dictionary, then rank of word "VIRAT" is :
 (A) 100 (B) 105 (C) 107 (D) 109
- If $\alpha, \beta, \gamma \in [0, 2\pi]$, then the sum of all possible values of α, β, γ .
 If $\sin \alpha = -\frac{1}{\sqrt{2}}, \cos \beta = \frac{-1}{2}, \tan \gamma = -\sqrt{3}$, is:
 (A) $\frac{22\pi}{3}$ (B) $\frac{21\pi}{3}$ (C) $\frac{20\pi}{3}$ (D) 8π
- Number of words that can be formed using all the letters of word "PARKAR" if alike letters of one kind are together but are separated from other set of alike letters:
 (A) 24 (B) 30 (C) 18 (D) 12
- A shelf contains 25 different books, 7 are in single volume and other are in set of 9, 6, 3 volumes respectively. Total number of ways in which books may be arranged on the shelf if volumes of each set are together and in increasing or decreasing order.
 (A) $10!$ (B) $8 \times 10!$ (C) $72 \times 9!$ (D) $12 \times 10!$
- There are counters available in 7 different colours the counter are alike except for colour. The total number of arrangement consisting 8 counters assuming sufficient no. of counter of each colour if no arrangement consist all counters of the same colour is :
 (A) $8^7 - 7$ (B) $8^7 - 8$ (C) $7^8 - 7$ (D) $7^8 - 8$
- The value of $\cos\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{8}\right)\cos\left(\frac{\pi}{16}\right)\dots\cos\left(\frac{\pi}{2^n}\right)$ equals:
 (A) $\frac{1}{2^n} \operatorname{cosec}\left(\frac{\pi}{2^n}\right)$ (B) $\frac{1}{2^{n-1}} \operatorname{cosec}\left(\frac{\pi}{2^{n-1}}\right)$
 (C) $\frac{1}{2^n} \operatorname{cosec}\left(\frac{\pi}{2^{n-1}}\right)$ (D) $\frac{1}{2^{n-1}} \operatorname{cosec}\left(\frac{\pi}{2^n}\right)$
- Shyam has 11 identical coins. He wants to give some of them to 3 beggars and keep the remaining for himself, then number of different ways to do so is (He may keep all of them):
 (A) 220 (B) 286 (C) 364 (D) 455
- The number of positive integral solutions of the equation $x_1 x_2 x_3 = 150$ is:
 (A) 54 (B) 27 (C) 81 (D) None of these
- There are 9 pairs of shoes in rack. Four shoes are selected at random then number of ways so that at least one pair of shoes is formed:
 (A) 360 (B) 1260 (C) 1044 (D) 1126

10. The number of words of 6 letters containing equal number of vowels and consonants (repetition allowed) is:
 (A) $20 \times (105)^2$ (B) $10 \times (105)^3$ (C) $105^2 \times 6$ (D) $20 \times (105)^3$
11. 11 persons are made to sit in a row, then number of ways of selecting of 4 persons such that no two of them are consecutive is:
 (A) 8C_4 (B) ${}^{10}C_4$ (C) ${}^{11}C_4$ (D) ${}^{13}C_4$
12. Let $S = \sin 2 - \sin 3$ and $C = \cos 2 - \cos 3$, then which one of the following is correct?
 (A) $S > 0$ and $C > 0$ (B) $S > 0$ and $C < 0$
 (C) $S < 0$ and $C > 0$ (D) $S < 0$ and $C < 0$
13. Let $a + b = 3 - \cos 4\theta$ and $a - b = 4 \sin 2\theta$, then maximum value of 'ab' is equal to:
 (A) $\frac{1}{2}$ (B) 1 (C) $\frac{2}{3}$ (D) 3
14. In a triangle ABC , angle $C < \text{angle } B$, angle B and angle C satisfy the equation $2 \tan x - \alpha(1 + \tan^2 x) = 0$ (where $0 < \alpha < 1$). Then the measure of angle A is:
 (A) $\frac{\pi}{3}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$
15. A student is to answer 10 out of 13 questions in an examination such that he must choose atleast 3 from the first five questions. The number of choices available to him is:
 (A) 140 (B) 196 (C) 276 (D) 346
16. A shopkeeper sells four varieties of perfumes and he has a large number of bottles of the same size of each variety in his stock. There are 6 places in row in his showcase. The number of different ways of displaying the four varieties of perfumes in the showcase is:
 (A) 1240 (B) 1560 (C) 150 (D) 960
17. Total number of solution(s) of the equation $\sin^4 2x + \cos^4 2x = \sin 2x \times \cos 2x$ in $(0, \pi)$ is :
 (A) 1 (B) 2 (C) 3 (D) 4
18. 10 different toys have to be distributed among 10 children. Total number of ways in which these toys can be distributed so that exactly one child gets no toy, is equal to:
 (A) $10!$ (B) $10! {}^{10}C_2$ (C) $9! {}^{10}C_2$ (D) $10! {}^9C_2$
19. The solution of the equation $\frac{\sin 3x}{2 \cos 2x + 1} = \frac{1}{2}$ is:
 (A) $x = 3n\pi + \frac{\pi}{6}$ (B) $x = n\pi + \frac{\pi}{6}$
 (C) $x = n\pi + (-1)^n \frac{\pi}{3}$ (D) $x = n\pi + (-1)^n \frac{\pi}{6}$
20. There are 12 points in plane of which 5 are collinear then maximum number of distinct triangles that can be formed with vertices at these points:
 (A) 210 (B) 310 (C) 410 (D) (510)

SPACE FOR ROUGH WORK

SECTION-2

This Section contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 999 (both inclusive).

1. Number of solutions for equation $2\cos^2 x = \sin x + 1$ in $[0, 4\pi]$ is:
2. If $\frac{\sin^4 \theta}{5} + \frac{\cos^4 \theta}{1} = \frac{1}{6}$, then $\sec^2 \theta$ is equal to:
3. Number of 4 letter word using the letters of word "MAHABHARAT" is 'a', then remainder when a is divided by 100 is:
4. Number of divisors of $2^5 \cdot 3^3 \cdot 5^5 \cdot 6^3$ is:
5. There are 3 set of distinct parallel lines containing 6, 5, 4 lines respectively. Find the maximum number of possible parallelograms formed by their points of intersection.

SPACE FOR ROUGH WORK